2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)
16 – 18 April 2026, Chittagong, Bangladesh

2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN) | 979-8-3315-4990-9/26/$31.00 ©2026 IEEE | DOI: 10.1109/QPAIN69676.2026.11546202

Design of an Area-Efficient and High-Speed 8-Bit Hybrid Carry Select Adder for Low-Power VLSI Applications

Sanuwarul Islam
Department of Electrical, Electronic and Communication Engineering
Military Institute of Science and Technology, Bangladesh
zimsanuwar@gmail.com

Tanzid Ahmed
Department of Electrical, Electronic and Communication Engineering
Military Institute of Science and Technology, Bangladesh
tanzidahmed29@gmail.com

Nazmus Sakib Rafi
Department of Electrical, Electronic and Communication Engineering
Military Institute of Science and Technology, Bangladesh
nsrafi85@gmail.com

Md. Faiaz Shajid
Department of Electrical, Electronic and Communication Engineering
Military Institute of Science and Technology, Bangladesh
faiazshajid1234@gmail.com

***Abstract*— In this work, an area efficient high-speed 8-bit hybrid carry select adder (CSA) is designed and implemented using CMOS, TGL & PTL technology. The proposed architecture integrates a 4-bit carry lookahead adder (CLA) for the least significant bits and two parallel 4-bit ripple carry adders (RCA) with a 2×1 multiplexer for the most significant bits, delivering a design that balances fast operation with reduced area and manageable circuitry. By leveraging the fast carry computation of CLA and the simplicity of RCA, the hybrid design significantly reduces overall propagation delay and rise-time delay compared to conventional ripple carry and standard CSA designs. The novelty of this work lies in the heterogeneous logic integration of CMOS, TGL, and PTL within a single CSA architecture to achieve reduced delay and transistor count. Performance evaluation across multiple design parameters demonstrates that the proposed 8-bit CSA effectively minimizes delay while maintaining less transistor, making it suitable for high-performance and low-power digital applications.**

Keywords*— *Carry Select Adder (CSA), Ripple Carry Adder (RCA), Carry Look-Ahead Adder (CLA), Low Power,High- Speed, CMOS Technology, Multiplexer (MUX)

I. Introduction

Addition is the most frequently employed arithmetic function in digital electronics, acting as a foundational element in processors and computational systems, serving as the basis for arithmetic logic units (ALUs), microprocessors, and signal processing units. Since adders are used extensively in computing architectures, the efficiency of an adder directly influences the overall performance, power consumption, and silicon area of modern integrated circuits [1]. With the growing demand for high-speed and low-power digital systems, researchers have continuously explored different adder architectures to optimize these conflicting design parameters. Among the most widely studied designs are the Ripple Carry Adder (RCA), the Carry Lookahead Adder (CLA), and the each of which provides unique advantages and limitations [2]–[4].

The RCA is the simplest and most area-efficient architecture, consisting of cascaded 1-bit full adders where the carry output of previous stage feeds into the next. Its hardware simplicity makes it attractive for low-cost systems; however, its major drawback lies in the linear carry propagation delay, which increases proportionally with bit-width and significantly limits performance in wide adders [5].

To overcome this limitation, the CLA was developed. The CLA improves speed because the carry here doesn't depend on the previous computation and carry out can generate in parallel, thereby reducing the dependency on previous carry [6]. While CLA achieves much faster operation than RCA, its wiring complexity and increased logic overhead limit its scalability in larger architectures, especially beyond 16 or 32 bits, where routing congestion becomes significant [7].

The Carry Select Adder provides another approach by reducing propagation delay through parallel precomputation. In this method, two adder work in parallel, having carry-in values of both '0' and '1' respectively, and a MUX is used to select the correct adder once the actual carry is known [8]. Although CSA achieves better performance than RCA and offers a simpler structure than CLA, its main drawback is redundant hardware utilization, which increases area and power consumption. To address this, several optimizations have been proposed, such as replacing one RCA with a Binary Excess-1 Converter (BEC), which reduces hardware duplication while maintaining speed improvements [9], [10].

In recent years, hybrid adder architectures have emerged as promising solutions, combining the strengths of RCA, CLA, and CSA while mitigating their respective weaknesses. These designs strategically integrate different logic blocks within the same architecture—such as employing RCA for less significant bits, CLA for middle stages, and CSA with BEC optimization for higher bits—thereby achieving a balanced trade-off between delay, area, and power [11], [12]. Such hybrid approaches are particularly valuable in low-power VLSI design, where efficiency is as critical as computational speed, especially in portable devices, embedded systems, and energy-constrained IoT applications [13].

In this work, we propose an optimized 8-bit hybrid Carry Select Adder that leverages the strengths of RCA, CLA, and CSA architectures. Our design reduces the propagation delay problem inherent in RCA, minimizes the wiring and complexity challenges of CLA, and eliminates redundant area overhead in traditional CSA. We used TGL for the CLA to speed up carry computation and reduce power, and PTL for the MUX to make selection faster and save area.

Unlike conventional BEC-based and homogeneous CSA implementations that rely solely on architectural restructuring, the

979-8-3315-4990-9/26/$31.00 ©2026 IEEE

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:50 UTC from IEEE Xplore. Restrictions apply.

proposed design introduces heterogeneous logic integration at the circuit level. CMOS logic is used for the CLA, Transmission Gate Logic (TGL) for the RCA blocks, and Pass Transistor Logic (PTL) for multiplexers.

II. Methodology

The 8-bit CSA is designed to perform high-speed addition of two 8-bit binary numbers (A0-A7 and B0-B7) with a carry- in (Cin). Here we used the advantages of both CLA and RCAs to minimize delay and optimize power consumption. This gives us a balance between speed, power, and area, making the CSA suitable for high-performance applications.

This process is done in three steps:

A. Carry Look-Ahead Adder using CMOS

The CLA is used for the addition of the least significant four bits (LSB) of the input operand (A_0-A_3) and (B_0-B_3) along with the carry-in (C_{in}) and is implemented using Complementary Metal-Oxide- Semiconductor (CMOS) technology for high-speed operation. The CLA operates by precomputing carry signals using the propagate ($P_i=A_i \oplus B_i$) and generate ($G_i = A_i.B_i$) functions, which help in calculating a carry at each bit position without dependent on previous calculation. The carry signals are computed in parallel using the equations,

$$C_1 = G_0 + (P_0 C_{in}),$$
$$C_2 = G_1 + (P_1.C_1)$$
$$C_3 = G_2 + (P_2.C_2)$$
$$C_4 = G_3 + (P_3.C_3)$$

thereby reducing the propagation delay significantly. The sum bits(S_0-S_3) are obtained using the arithmetic XOR operation ($P_i=A_i \oplus B_i$). The entire CLA circuit is realized using CMOS logic gates, where PMOS and NMOS transistors implement AND, OR, and XOR operations, ensuring optimal power efficiency and reduced delay. CMOS-based CLA design offers several advantages, including high-speed computation due to parallel carry generation, low power consumption by minimizing static power dissipation, and scalability for integration into larger digital circuits. The final carry-out (C_4) from the CLA is utilized as a selection signal for the multiplexers in the subsequent RCA stage, which determines the sum output for the higher-order bits (S_4-S_7) . Though CLA is fast it needs more area.

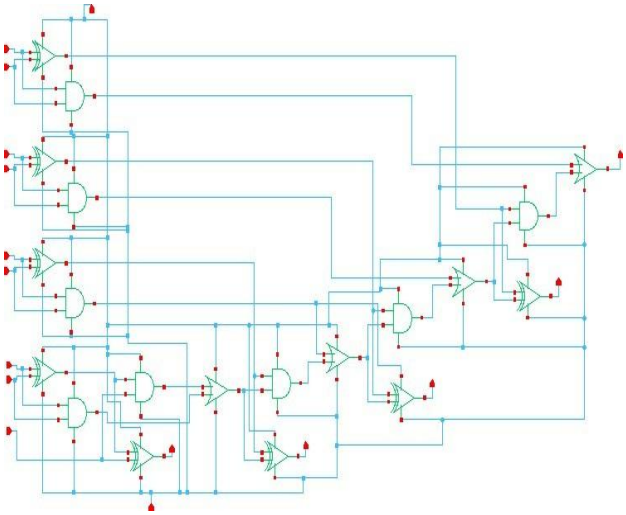


Figure 1:Carry Look-Ahead Adder

B. Ripple Carry Adder using Transmission Gate Logic

The RCA is used to compute the sum of the higher-order bits *(A_4-A_7 and B_4-B_7)* and is implemented using Transmission Gate Logic (TGL), which is the novelty of our design. Unlike conventional static CMOS logic, TGL offers improved performance by reducing transistor count and power dissipation while maintaining full-swing voltage characteristics. Two separate RCAs are designed: one having an initial carry-in of 0 and the other assuming an initial carry-in of 1. Each RCA consists of a series of full adders, where the carry output of one stage propagates as the carry input to the next stage, following the equations

$$S_i = A_i \oplus B_i$$

$$C_{i+1} = (A_i.B_i),\big(C_i.(A_i \oplus B_i)\big).$$

RCA needs less area than CLA but it is slower compared to CLA.

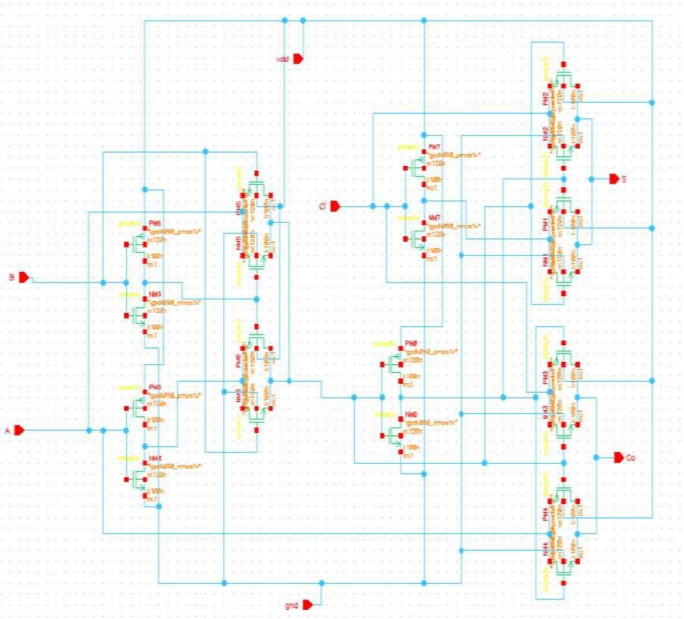


Figure 2: Ripple Carry Adder (1 bit)

The full adder cells within the RCA are implemented using TGL, which consist of parallel PMOS and NMOS transistors controlled by complementary signals to achieve low resistance and minimal voltage drop. This approach enhances switching speed and power efficiency compared to conventional CMOS pass-transistor logic. The primary advantages of using transmission gate logic in RCA implementation include reduced propagation delay due to lower resistance paths, minimal power dissipation by eliminating unnecessary static currents, and improved layout density, making it suitable for high-performance arithmetic circuits. The outputs of both RCAs, corresponding to sum bits (S_4-S_7). are then fed into multiplexers controlled by the carry-out from the CLA, selecting the correct sum output of RCA based on the actual carry value.

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:50 UTC from IEEE Xplore. Restrictions apply.

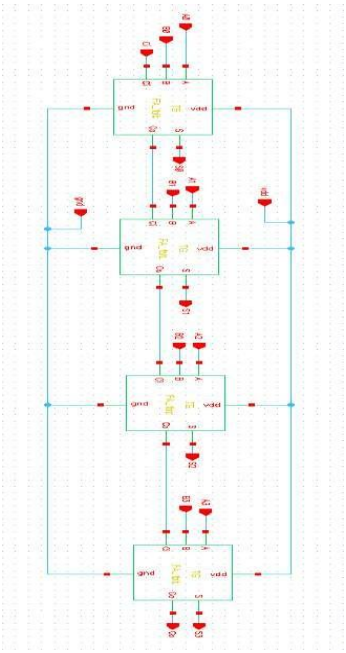


Figure 3:4-bit RCA

C. 2x1 MUX using Pass Transistor Logic

To determine the final sum output for the higher-order bits,*(S_4 - S_7)* we use a selection mechanism involving multiplexers (MUX), which are implemented using Pass Transistor Logic (PTL). The novelty of our design lies in utilizing PTL for all multiplexers, ensuring reduced transistor count, lower power consumption, and faster switching speeds compared to conventional CMOS-based MUX designs. Each sum bit *(S_4 - S_7)* is obtained by selecting between the outputs of the two Ripple Carry Adders (RCAs)—one assuming an initial carry-in of 0 and the other assuming an initial carry-in of 1. Specifically, each bit pair *(S_4, S_5, S_6, S_7)* from both RCAs is connected to a 2-to-1 multiplexer, where the selection signal is provided by the carry-out *(C_4)* of the CLA. If C_4=0 , the multiplexer selects the sum outputs from the RCA with an initial carry-in of 0; otherwise, it selects the outputs from the RCA with an initial carry-in of 1. Additionally, a separate 2-to-1 multiplexer is used to select the final carry- out by choosing between the carry-out signals of the two RCAs.

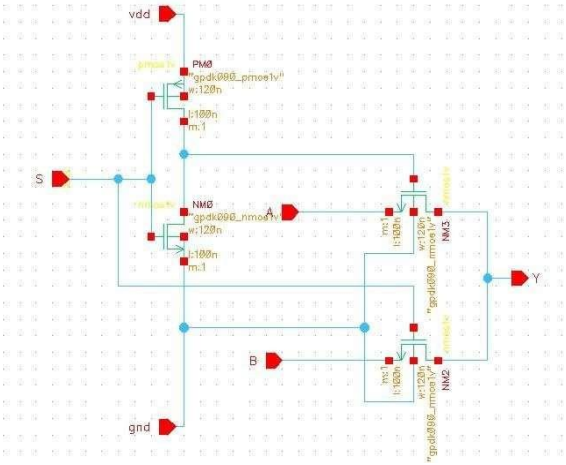


Figure 4:2x1 MUX using PTL

Using Pass Transistor Logic (PTL) for multiplexer implementation offers significant advantages over traditional CMOS logic. PTL reduces the number of transistors required per MUX, leading to a more compact design with improved area efficiency. Furthermore, PTL minimizes power dissipation by reducing the number of switching nodes and eliminating redundant voltage transitions, making it highly suitable for low-power arithmetic circuits. The reduced capacitance in PTL-based multiplexers also results in faster signal propagation, enhancing the overall speed of the Carry Select Adder. By integrating PTL-based MUX circuits with our transmission gate logic (TGL)-based RCA and CMOS-based CLA, our design achieves a highly efficient and optimized 8-bit CSA with improved performance in terms of speed, power, and area utilization.

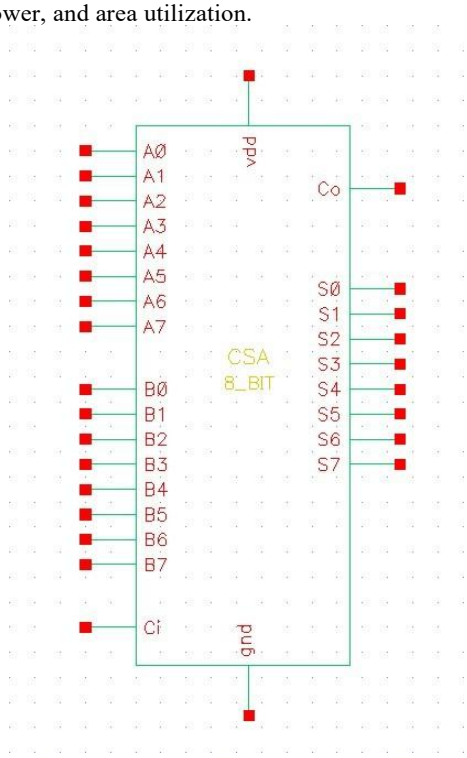


Figure 5:8-bit CSA symbol

As illustrated in Figure 6, the proposed 8-bit adder architecture processes the least significant 4 bits (LSB) of the input operands using a 4-bit Carry Lookahead Adder (CLA) to achieve fast carry computation. The most significant 4 bits (MSB) are handled by two parallel 4-bit Ripple Carry Adders (RCA), each assuming carry-in values of 0 and 1. A 2×1 multiplexer dynamically selects the appropriate RCA output based on the carry-out from the CLA, ensuring accurate and efficient sum generation. In this hybrid design, the RCAs are implemented using Transmission Gate Logic (TGL) for low-area, low-power operation, while the multiplexer is realized with Pass Transistor Logic (PTL) to minimize transistor count and switching delay. By combining CLA, RCA, and PTL-based multiplexing, the architecture reduces propagation delay, lowers rise-time, and optimizes power consumption compared to conventional CSA implementations.

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:50 UTC from IEEE Xplore. Restrictions apply.

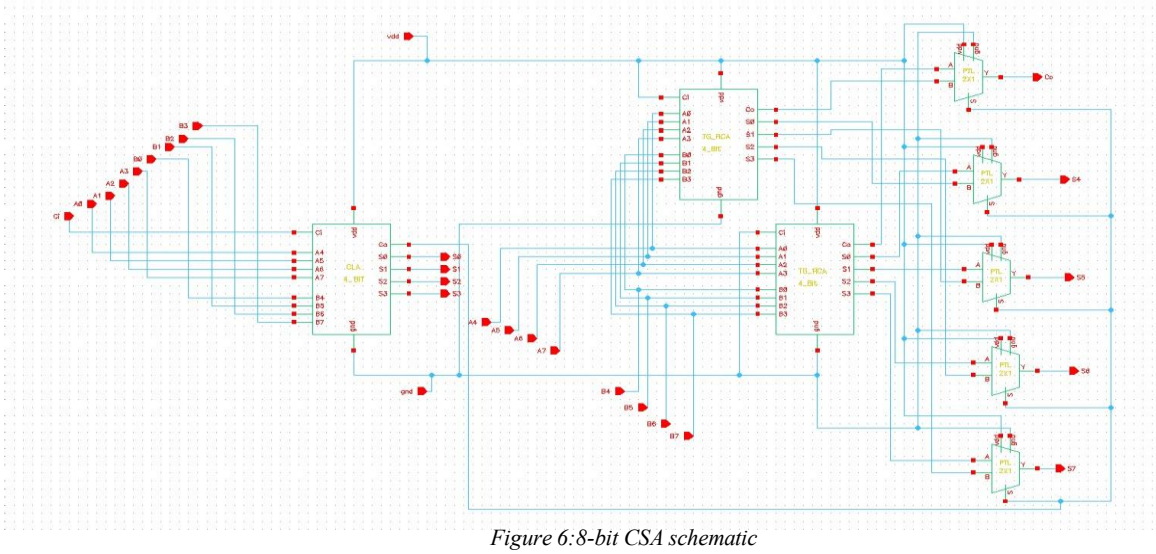
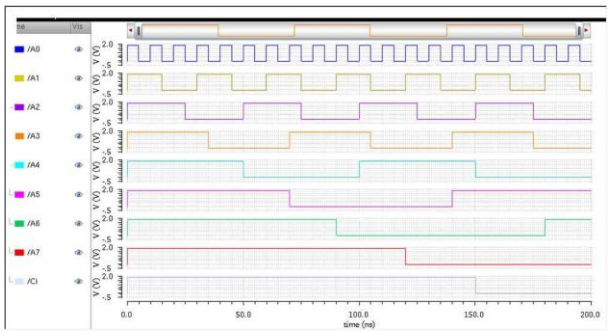


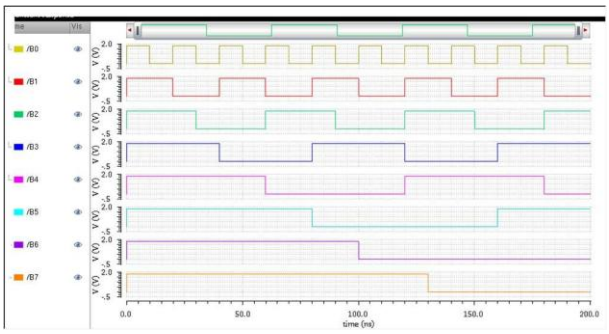
Figure 6:8-bit CSA schematic

III. Result and Findings

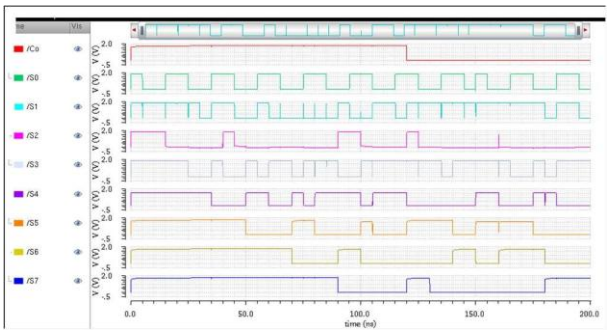
A. Input and Output waveforms of CSA



(a) Input A



(b) Input B



(c) Output

Figure 7: Input waveforms are shown in (a) & (b); Output waveform shown in (c)

B. Comparison between CSA and RCA rise time delay:

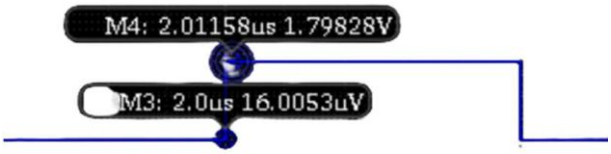


Figure 8(a): CSA rise time Delay

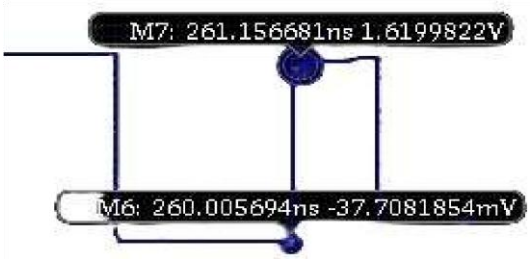


Figure 8(b): RCA rise time Delay

The input waveforms for operands A and B and the resulting CSA output are shown in Figures 7(a)-7(c). The output waveform confirms correct addition, demonstrating that the hybrid CSA accurately computes the sum for the given inputs.

The proposed 8-bit hybrid Carry Select Adder (CSA) has been evaluated in terms of rise-time, propagation delay, and transistor count. The architecture employs a 4-bit CLA for the least significant bits (LSB) and two parallel 4-bit RCAs for the most significant bits (MSB), with a 2×1 PTL multiplexer selecting the RCA output based on the CLA carry-out. This hybrid approach enables fast carry computation for LSB and parallel sum calculation for MSB, reducing overall delay and power consumption.

The rise-time delay, obtained from Figure 8, of the CSA output is calculated as:

$$t_{rise,CSA} = 261.156681\ \text{ns} - 260.005694\ \text{ns} \approx 1.151\ \text{ns}$$

For the conventional RCA, the rise-time delay is:

$$t_{rise,RCA} = 2.01158\ \mu\text{s} - 2.0\ \mu\text{s} \approx 11.58\ \text{ns}$$

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:50 UTC from IEEE Xplore. Restrictions apply.

C. Comparison between other adder architectures:

Architecture	Technology (GDPK)	Rise-Time Delay (ns)	Transistor Count	Remarks
8-bit RCA	90nm	11.58	224	Simple but slower
8-bit CLA	90nm	—	344	Parallel carry, higher area
Proposed Hybrid CSA	90nm	1.151	299	Faster than RCA, Fewer Transistor than CLA

The proposed hybrid CSA achieves a measured rise-time delay of 1.151 ns compared to 11.58 ns for RCA, corresponding to approximately 90% delay reduction. With 299 transistors, it provides a balanced trade-off between the compact RCA and the area-intensive CLA. Prior literature primarily focuses on architectural optimization, whereas this work emphasizes circuit-level heterogeneous logic integration.

In summary, the hybrid CSA achieves a balanced trade-off between speed, area, and power, outperforming conventional RCA and CLA designs, making it suitable for low-power, high-performance digital systems.

IV. Conclusion

This paper presents a high-speed, low-power 8-bit hybrid CSA optimized for modern VLSI applications. The proposed design combines a 4-bit CLA for the least significant bits with two parallel 4-bit Transmission Gate (TG)-based RCA for the most significant bits, and a Pass Transistor Logic (PTL) multiplexer for efficient carry selection. This hybrid architecture leverages the speed of CLA, the low-power and area efficiency of TG-based RCA, and the reduced transistor count of PTL multiplexers, resulting in improved performance over conventional RCA and CLA designs. Simulation results demonstrate accurate addition with significant reductions in propagation delay, rise-time, and transistor utilization, confirming the effectiveness of the approach. The design achieves a balanced trade-off among speed, power, and area, making it highly suitable for high-performance, low-power digital systems. Future work includes layout implementation to facilitate fabrication and practical evaluation.

V. Simulation Environment

All circuits were implemented in Cadence Virtuoso using GPDK 90 nm CMOS technology at VDD = 1.0 V and room temperature (27°C). Transient analysis was performed to verify functionality and measure rise-time delay under identical input conditions for RCA, CLA and the proposed hybrid CSA.

References

[1] S. Parmar and K. P. Singh, "Design of high speed hybrid carry select adder," in *2013 3rd IEEE International Advance Computing Conference (IACC)*, Ghaziabad, 2013, pp. 1656–1663, doi: 10.1109/IAdCC.2013.6514477.

[2] P. Balasubramanian and N. E. Mastorakis, "High-Speed and Energy-Efficient Carry Look-Ahead Adder," *Journal of Low Power Electronics and Applications*, vol. 12, no. 3, p. 46, Aug. 2022, doi: 10.3390/jlpea12030046.

[3] "A Review On N-Bit Ripple-Carry Adder, Carry-Select Adder And Carry-Skip Adder," *Journal of VLSI Circuits and Systems*, vol. 4, no. 1, Jan. 2023, doi: 10.31838/jvcs/04.01.05.

[4] S. M. Sulaiman, I. Jaiswal, I. Rawat, and J. Belani, "Design and Implementation of 4 Bit Carry Generator Circuit Using Pass Transistor Logic with 2 × 1 Mux Approach," in *2024 International Conference on Communication, Computer Sciences and Engineering (IC3SE)*, Gautam Buddha Nagar, India, May 2024, pp. 1–5, doi: 10.1109/IC3SE62002.2024.10593549.

[5] V. G. Oklobdzija, D. Villeger, and M. S. K. Choi, "Area, Delay and Power Comparison of Adder Topologies," VLSI Circuits and Systems (VLSICS), 2012, pp. 1–6, doi: 10.1109/VLSICS.2012.6225698..

[6] A. K. Panda, R. Palisetty, and K. C. Ray, "High-Speed Area-Efficient VLSI Architecture of Three-Operand Binary Adder," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 67, no. 11, pp. 3944–3953, Nov. 2020, doi: 10.1109/TCSI.2020.3016275.

[7] M. Tammana, M. Vardhan, G. R. S., A. S., and A. Anujith, "Efficiency Analysis of Adder Architectures: A Comparative Study Across Various Bit Lengths," in *2024 IEEE International Conference on Information Technology, Electronics and Intelligent Communication Systems* (ICITEICS), Bangalore, India, Jun. 2024, pp. 1–6, doi: 10.1109/ICITEICS61368.2024.10625527.

[8] C.-J. Fang, C.-H. Huang, J.-S. Wang, and C.-W. Yeh, "Fast and compact dynamic ripple carry adder design," in Proceedings. IEEE Asia-Pacific Conference on ASIC, Taipei, Taiwan, 2002, pp. 25–28, doi: 10.1109/APASIC.2002.1031523.

[9] G. R. Padmini, O. Rajesh, K. Raghu, N. M. Sree, C. Apurva, and K. Saikumar, "Design and Analysis of 8-bit ripple Carry Adder using nine Transistor Full Adder," in *2021 7th International Conference on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India, Mar. 2021, pp. 1982– 1987, doi: 10.1109/ICACCS51430.2021.9441928.

[10] B. K. Mohanty and S. K. Patel, "Area–Delay–Power Efficient Carry-Select Adder," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 61, no. 6, pp. 418–422, Jun. 2014, doi: 10.1109/TCSII.2014.2319695.

[11] P. Balasubramanian and N. Mastorakis, "Performance Comparison of Carry-Lookahead and Carry-Select Adders Based on Accurate and Approximate Additions," *Electronics*, vol. 7, no. 12, p. 369, Dec. 2018, doi: 10.3390/electronics7120369.

[12] J. L. Lakshmesha and K. R. Usha Rani, "A Novel Ripple/Carry Lookahead Hybrid Carry Select Adder Architecture," *International Journal of Computer Applications*, vol. 70, no. 27, pp. 5–9, May 2013, doi: 10.5120/12237-8416.

[13] [13] P. Swathi and G. B. Gupta, "Implementation of 2-bit Multiplier Circuit Using Pass Transistor Logic," *International Journal for Research in Applied Science and Engineering Technology*, vol. 10, no. 7, pp. 5013–5022, Jul. 2022, doi: 10.22214/ijraset.2022.46100.

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:50 UTC from IEEE Xplore. Restrictions apply.

Authorized licensed use limited to: INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY. Downloaded on October 25,2026 at 13:37:50 UTC from IEEE Xplore. Restrictions apply.